

K-Means Clustering Algorithm Based on GEO Optimization

Xu Zhang

*School of Electronics and Information
Northwestern Polytechnical University
Xi'an, China
zhangxu112@mail.nwpu.edu.cn*

Zesheng Dan*

*School of Electronics and Information
Northwestern Polytechnical University
Xi'an, China
zsdan@nwpu.edu.cn*

Yangyang Liu

*School of Electronics and Information
Northwestern Polytechnical University
Xi'an, China
yyliau@nwpu.edu.cn*

Deyan Li

*China United Network Communications
Co.,Ltd..Hainan Branch
Hainan, China
lidy136@chinaunicom.cn*

Xiaoting Zhang

*Unicom (Hainan) Industrial Internet
Co., Ltd.
Hainan, China.
zhangxt74@chinaunicom.cn*

Chengkai Tang

*School of Electronics and Information
Northwestern Polytechnical University
Xi'an, China
cktang@nwpu.edu.cn*

Corresponding author: Zesheng Dan (E-mail: zsdan@nwpu.edu.cn)

Abstract—This paper proposes a k-means algorithm enhanced with GEO (Golden Eagle Optimizer) optimization to address the challenges encountered by traditional k-means algorithm, such as being highly sensitive to initial values and prone to getting stuck in local optima during clustering. By incorporating GEO's attack and cruising vectors into the original loss function and iterative process of k-means, our algorithm enhances its exploratory capability while retaining its original clustering prowess. Theoretical analysis and simulation results demonstrate that our method can further minimize the loss function and exhibit superior clustering performance.

Index Terms—clustering algorithm, GEO algorithm, k-means algorithm, optimization, monte carlo simulation

I. INTRODUCTION

In the fields of data mining and optimization, clustering algorithms serve as powerful tools for uncovering hidden structures and patterns within datasets. By segmenting data based on similarity, these algorithms facilitate decision-making and predictive modeling by establishing clear boundaries. They find extensive applications in various domains, including task planning[1], image analysis[2], network exploration[3], and electrical engineering[4], as well as the field of multi-information fusion and data analysis[5–14].

The k-means algorithm, renowned for its simplicity, efficiency, and scalability, stands as a classical clustering method. However, the algorithm's inherent limitations often result in a non-convex solution space when determining the number of cluster centers[15, 16]. Specifically, the objective function of the k-means algorithm, aimed at minimizing the sum of squared errors (SSE) between each data point and its assigned

cluster centers, yields a non-convex function. Consequently, the algorithm may become ensnared in local optima, thereby influencing the clustering outcomes through the initial placement of cluster centers. To address this challenge, strategies such as multiple runs of the algorithm to select the optimal clustering result or enhancements to the k-means algorithm for improved clustering efficacy can be employed[17–21].

Hence, the key to improving the performance of the k-means algorithm is to prevent it from getting trapped in local optima, thereby attaining the global optimal solution. The GEO (Golden Eagle Optimizer) algorithm has demonstrated effectiveness in solving global optimization problems[22]. Inspired by the strategy of golden eagles adjusting their speed at different stages of their spiral trajectory, the GEO algorithm explores and develops the global area by adjusting flight directions at different stages, thus achieving global optimization.

In response to the k-means algorithm's susceptibility to local optima, this paper presents a GEO-optimized k-means clustering algorithm. Integrating the strengths of the GEO algorithm, this approach introduces attack vectors, cruising vectors, and random memory selection into the k-means iterative process. By adjusting the development strategies at various stages, it accomplishes a comprehensive global exploration. Experimental findings indicate that the proposed enhanced algorithm surpasses traditional methods, leading to a further reduction in the loss function and achieving superior clustering quality and global search capabilities.

II. TRADITIONAL K-MEANS ALGORITHM AND GEO ALGORITHM.

A. K-Means Algorithm

When handling large datasets, the k-means algorithm offers the advantages of rapid convergence and superior clustering performance. Its fundamental concept involves itera-

tively seeking a partitioning scheme of K clusters within the data[23], aiming to minimize the loss function associated with the clustering outcomes. This loss function is expressed as follows:

$$J(c, \mu) = \sum_{i=1}^M \|x_i - \mu_{c_i}\|^2 \quad (1)$$

Where: x_i represents the i -th sample, c_i represents the cluster to which x_i belongs, μ_{c_i} represents the centroid corresponding to cluster i , and M is the total number of samples.

The specific steps are as follows:

- a) Data preprocessing, including standardization and outlier filtering.
- b) Selecting k cluster centers $\mu_{c_1}, \mu_{c_2}, \dots, \mu_{c_k}$, denoted as U .
- c) Defining the loss function $J(c, \mu)$.
- d) Let t denote the number of iterations. Repeat the following process until $J(c, \mu)$ converges:
 - aa) Assign each sample to the nearest cluster centers.

$$c_i^t < - \operatorname{argmin}_k \|x_i - \mu_k^t\|^2 \quad (2)$$

- bb) For each cluster centers k , recalculate the center position of that class.

$$\mu_k^{(t+1)} < - \operatorname{argmin}_\mu \sum_{i:c_i^t=k}^b \|x_i - \mu\|^2 \quad (3)$$

The core of k-means involves first fixing the cluster centers and adjusting the category of each sample to reduce $J(c, \mu)$, then fixing the category of each sample and adjusting the cluster centers to further minimize $J(c, \mu)$. These two processes alternate in a loop, with $J(c, \mu)$ monotonically decreasing until it reaches the minimum value, and both the cluster centers and the sample category assignments converge simultaneously. The iterative process is illustrated in Fig.1 below.

The evaluation criterion for assessing the results of the k-means algorithm is the size of the loss function, which represents the minimized sum of squared errors (SEE) between each data point and its assigned cluster centroid (refer to Fig.2). To enhance the iteration speed, it can be considered that clustering is finalized when the SEE experiences negligible changes.

B. GEO Algorithm

The GEO (Golden Eagle Optimizer) algorithm is a nature-inspired heuristic algorithm that mimics the predatory behavior of golden eagles. As illustrated in Fig.3, the motion direction of the golden eagle is composed of attack vectors and cruise vectors.

Where the attack vector, as shown in (4), points from the current position of the golden eagle to the target position:

$$\vec{A} = \vec{X}_P - \vec{X}_L \quad (4)$$

In the equation, $\vec{A}$ represents the attack vector, $\vec{X}_P$ denotes the position of the target point, and $\vec{X}_L$ stands for the current position of the golden eagle.

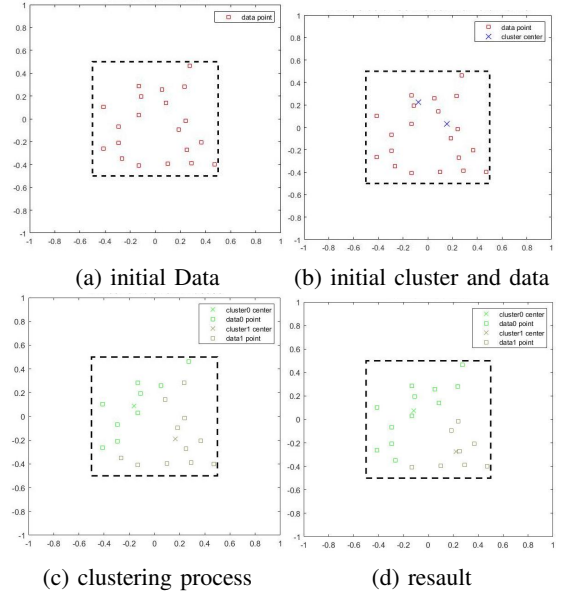


Fig. 1. The process of the k-means algorithm

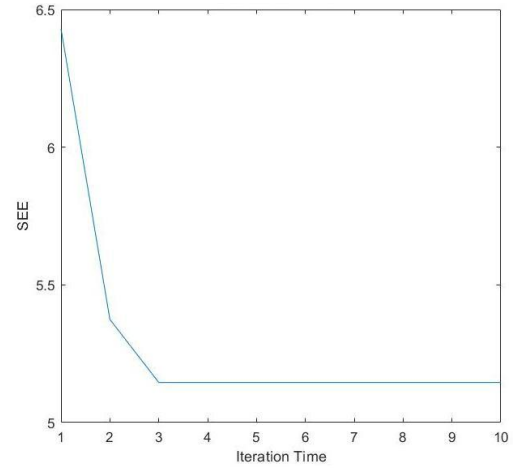


Fig. 2. k-means SEE

The cruise vector lies within the hyperplane containing the current position point of the golden eagle, with the attack vector as the normal vector. Hence, the attack vector and the cruise vector are orthogonal, and we can readily determine the expression for the hyperplane equation, which can be represented as (5).

$$\sum_{i=1}^n a_i x_i = d \quad (5)$$

Where a_i represents the component corresponding to the normal vector, and x_i stands for the variable.

Let the cruise vector be denoted as $\vec{C} = [c_1, c_2, \dots, c_n]$. According to the hyperplane equation where the cruise vector lies, we select the component c_k corresponding to the k -th non-zero normal vector a_k as the variable to be determined, while

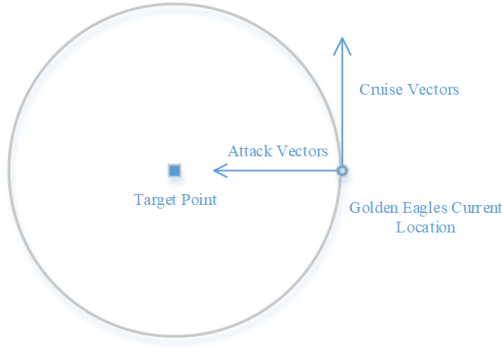


Fig. 3. The movement composition of the golden eagle.

the components of the remaining cruise vectors are assigned random values. From (5), we can obtain:

$$c_k = \frac{d - \sum_{i, i \neq k} a_i c_i}{a_k} \quad (6)$$

After obtaining the attack and cruise vectors, GEO progressively explores through iterations. During a complete iteration process of T iterations, the step size of the golden eagle's movement can be expressed as (7).

$$\Delta x = \vec{r_1} p_a \frac{\vec{A}}{\|\vec{A}\|} + \vec{r_2} p_c \frac{\vec{C}}{\|\vec{C}\|} \quad (7)$$

Where $\vec{r_1}$ and $\vec{r_2}$ are random vectors within the range $[0,1]$. p_a and p_c represent the attack coefficient and cruise coefficient, which can be adjusted to control the exploration step size by setting initial values p_a^0 and p_c^0 and final values p_a^T and p_c^T before the start of iteration.

$$p_a = p_a^0 + \frac{t}{T} |p_a^T - p_a^0| \quad (8)$$

$$p_c = p_c^0 - \frac{t}{T} |p_c^T - p_c^0| \quad (9)$$

Therefore, the position of the golden eagle at the t -th iteration can be represented as:

$$x^{t+1} = x^t + \Delta x^t \quad (10)$$

As shown in Fig.4, by adjusting the values of the attack coefficient and cruise coefficient, the golden eagle transitions from initially being predominantly cruise-oriented to gradually becoming predominantly attack-oriented, ultimately completing the exploration process.

III. K-MEANS CLUSTERING ALGORITHM BASED ON GEO OPTIMIZATION.

As seen in the previous section, the GEO algorithm achieves exploration by moving the position of the golden eagle. Therefore, as depicted in Fig.5, we consider the clustering performance of k-means as the objective to be optimized. We treat the number of k-means clusters as the number of golden

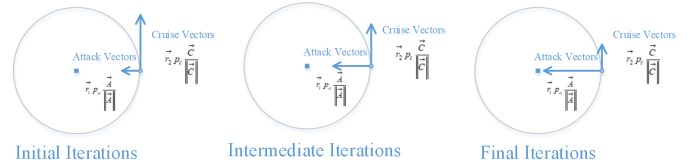


Fig. 4. The golden eagle gradually transitions from a cruise-oriented tendency to an attack-oriented tendency.

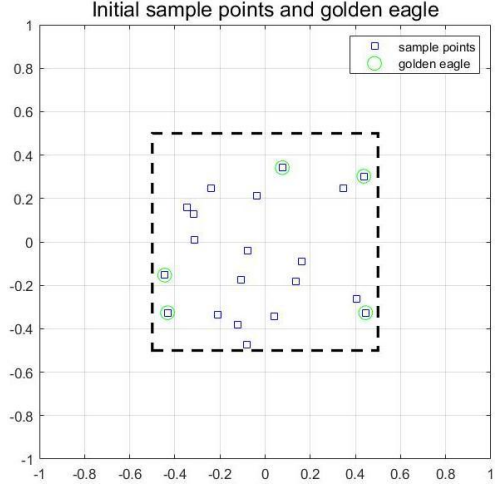


Fig. 5. Initial sample points and positions of the golden eagles.

eagle groups and assign initial positions to these groups based on the initial values of the k-means cluster centers.

When we place the groups of golden eagles in the data space that k-means needs to process, this exploration process can help k-means find the optimal positions for cluster centers, thus avoiding the problem of getting trapped in local optima and optimizing the entire clustering process.

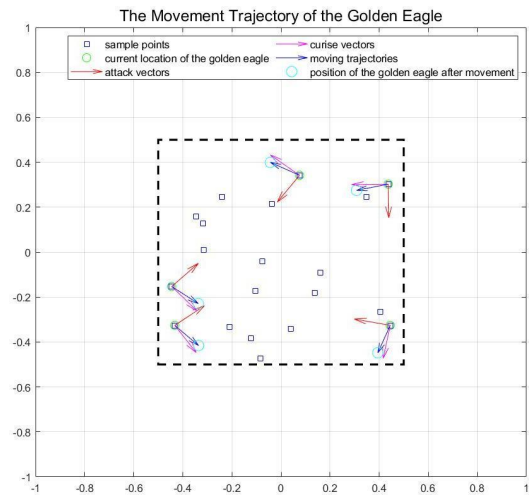


Fig. 6. The movement of golden eagles in the improved algorithm.

As shown in Fig.6, during iteration, the golden eagles

always explore towards their target positions. Although the positions of the golden eagles change with iteration, the entire group of golden eagles still retains the memory of their best-explored positions. In this algorithm, this corresponds to the position that minimizes the SEE for k-means, which we refer to as the optimal cluster center position. As the exploration process continues, if the clustering effect at the new position is better with a smaller SEE, the memory of the best position is updated.

In this algorithm, to increase the search space, we also incorporate a process of randomly exchanging population memories into the GEO algorithm(Fig.7). This helps reduce the limitations of the exploration process caused by the different initial positions of the algorithm.

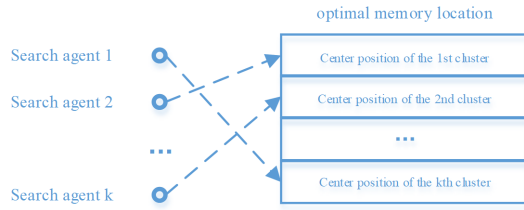


Fig. 7. The golden eagles randomly select the best target positions for exploration.

Based on the process described above, Algorithm 1 provides the pseudocode for the k-means clustering algorithm based on GEO optimization.

Algorithm 1 The pseudocode for the k-means clustering algorithm based on GEO optimization

- 1: Initialize the population of golden eagles.
- 2: Initialize the attack and cruise parameters p_a and p_c .
- 3: Initialize the population memory of the best target positions of golden eagles.
- 4: Initialize the number of iterations.
- 5: Calculate the initial k-means clustering performance based on the initial positions ((1) to (2)).
- 6: **for** Each iteration **do**
- 7: The golden eagles randomly select the best memory position from the population as the target.
- 8: Calculate the attack vector $\vec{A}$ (4).
- 9: **if** The attack vector is not equal to zero. **then**
- 10: Calculate the equation of the hyperplane where the cruise vector lies (5).
- 11: Calculate the cruise vector $\vec{C}$ (6).
- 12: **end if**
- 13: Calculate the step length for movement ((7) to (9)).
- 14: Update the position of the golden eagles (10).
- 15: **for** Combinations of new and old positions. **do**
- 16: Calculate the clustering performance of k-means ((1) to (3)).
- 17: **if** The current combination's SEE is smaller than the best position SEE memorized by the golden eagle group. **then**
- 18: Update the current cluster centers and the SEE of clustering to the best memory of the golden eagle group.
- 19: **end if**
- 20: **end for**
- 21: **end for**

IV. ALGORITHM SIMULATION RESULTS AND ANALYSIS.

In the simulation, 2D k-means data points were used for testing. The attack and cruise parameters were selected as $[p_a^0, p_a^T] = [0.25, 1]$ and $[p_c^0, p_c^T] = [0.5, 0.25]$ respectively, with the number of clusters specified as 5 and the number of samples as 20. The k-means++ method was used to select the initial points for clustering, and the initial positions for GEO golden eagles. The clustering results obtained by the proposed improved algorithm and the k-means++ method are shown in Fig.8 and Fig.9.



Fig. 8. The clustering results obtained using the k-means++ method.

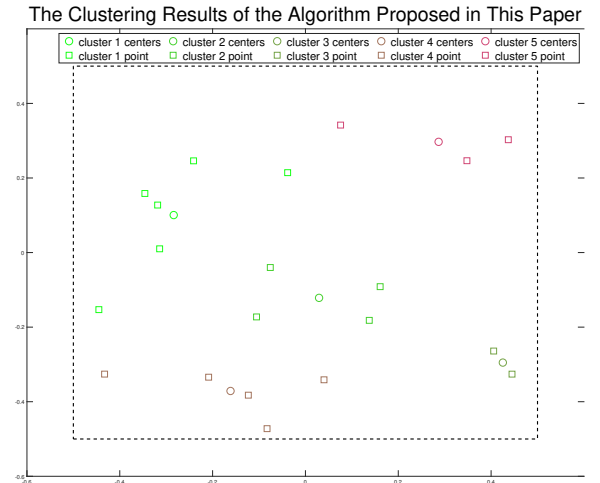


Fig. 9. The clustering results obtained using the proposed algorithm.

By comparing the Sum of Squared Errors (SEE) during the clustering process (Fig.10), we can clearly see that the proposed algorithm, through its exploration process, breaks out of local optima and finds more effective cluster centers, demonstrating stronger optimization capabilities.

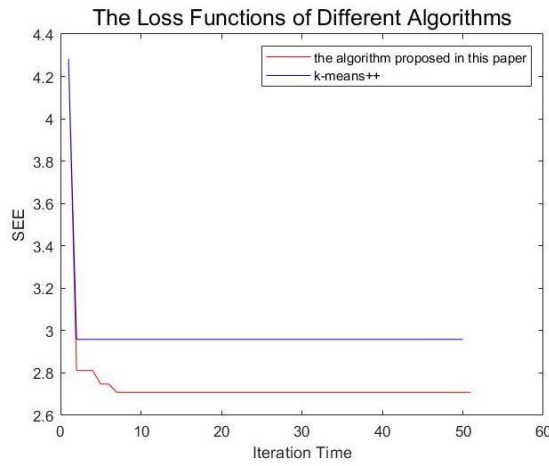


Fig. 10. SEE for the proposed algorithm and k-means++ clustering results.

To test the effectiveness of the algorithm, a monte carlo simulation experiment[24] was conducted for 1000 iterations. The experimental results are shown in Fig.11 and Fig.12.

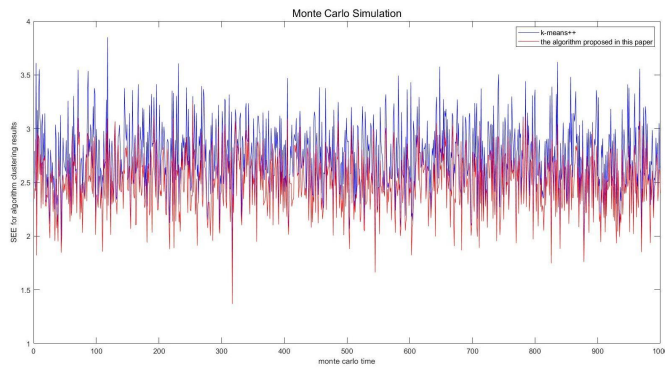


Fig. 11. Monte carlo simulation experiment results.

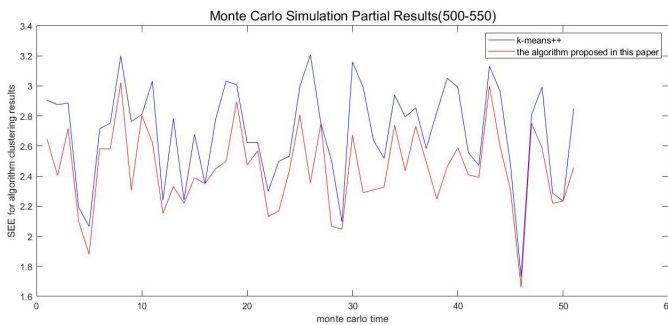


Fig. 12. Monte carlo simulation experiment results(partial).

During the Monte Carlo simulation experiment, the Sum of Squared Errors (SEE) of the clustering results obtained by the algorithm proposed in this paper and the SEE obtained by the k-means++ algorithm were recorded. From the results, it can be observed that the algorithm proposed in this paper further explores the overall data beyond the clustering limit achieved

by k-means. If the k-means algorithm has found the optimal solution, then the proposed algorithm can also derive the same optimal solution. However, if the k-means algorithm gets stuck in a local optimum, the proposed algorithm can further find a better solution.

V. SUMMARIZE

The paper introduces a GEO-optimized k-means clustering algorithm, which incorporates the exploration process of the GEO algorithm into the k-means algorithm. This enhancement enables further exploration of the dataset to reduce the limitation of the k-means algorithm frequently getting stuck in local optima. Simulation results demonstrate that the proposed algorithm exhibits superior optimization capability and faster convergence compared to the traditional k-means++ algorithm, making it a more efficient option.

REFERENCES

- [1] M. Elango, S. Nachiappan, and M. K. Tiwari, "Balancing task allocation in multi-robot systems using k-means clustering and auction based mechanisms," *Expert Systems with Applications*, vol. 38, no. 6, pp. 6486–6491, 2011.
- [2] N. Dhanachandra, K. Manglem, and Y. J. Chanu, "Image segmentation using k-means clustering algorithm and subtractive clustering algorithm," *Procedia Computer Science*, vol. 54, pp. 764–771, 2015.
- [3] S. Datta, C. Giannella, and H. Kargupta, "K-means clustering over a large, dynamic network," in *Proceedings of the 2006 SIAM international conference on data mining*, pp. 153–164, SIAM, 2006.
- [4] G. Zhang, Y. Li, and X. Deng, "K-means clustering-based electrical equipment identification for smart building application," *Information*, vol. 11, no. 1, p. 27, 2020.
- [5] Y. Yu, Y. Zhang, Z. Cheng, Z. Song, and C. Tang, "Mca: Multidimensional collaborative attention in deep convolutional neural networks for image recognition," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 107079, 2023.
- [6] Y. Yu, Y. Zhang, Z. Cheng, Z. Song, and C. Tang, "Multi-scale spatial pyramid attention mechanism for image recognition: An effective approach," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108261, 2024.
- [7] C. Tang, H. Shi, and L. Zhang, "Trajectory planning aided unmanned surface vehicle optimization communication method with hierarchical reinforcement learning," *Ocean Engineering*, vol. 307, p. 118225, 2024.
- [8] C. Tang, C. Wang, L. Zhang, Y. Zhang, and H. Song, "Vehicle heterogeneous multi-source information fusion positioning method," *IEEE Transactions on Vehicular Technology*, 2024.
- [9] L. Zhang, S. Wu, and C. Tang, "Cooperative positioning of underwater unmanned vehicle clusters based on factor graphs," *Ocean Engineering*, vol. 287, p. 115854, 2023.

- [10] C. Tang, H. Shi, and L. Zhang, "Geomagnetic matching cooperative positioning method for unmanned boat cluster based on factor graph," *Ocean Engineering*, vol. 296, p. 116901, 2024.
- [11] L. Zhang, S. Wu, and C. Tang, "Cooperative positioning of underwater unmanned vehicle clusters based on factor graphs," *Ocean Engineering*, vol. 287, p. 115854, 2023.
- [12] C. Tang, C. Wang, L. Zhang, Y. Zhang, and H. Song, "Multivehicle 3d cooperative positioning algorithm based on information geometric probability fusion of gnss/wireless station navigation," *Remote Sensing*, vol. 14, no. 23, p. 6094, 2022.
- [13] Z. Song, Y. Zhang, Y. Yu, and C. Tang, "Cooperative positioning algorithm based on manifold gradient filtering in uav-wsn," *IEEE Sensors Journal*, 2024.
- [14] C. Tang, H. Shi, and L. Zhang, "Geomagnetic matching cooperative positioning method for unmanned boat cluster based on factor graph," *Ocean Engineering*, vol. 296, p. 116901, 2024.
- [15] M. Ahmed, R. Seraj, and S. M. S. Islam, "The k-means algorithm: A comprehensive survey and performance evaluation," *Electronics*, vol. 9, no. 8, p. 1295, 2020.
- [16] A. Kapoor and A. Singhal, "A comparative study of k-means, k-means++ and fuzzy c-means clustering algorithms," in *2017 3rd international conference on computational intelligence & communication technology (CICT)*, pp. 1–6, IEEE, 2017.
- [17] Y.-K. Lam, P. W.-M. Tsang, and C.-S. Leung, "Pso-based k-means clustering with enhanced cluster matching for gene expression data," *Neural Computing and Applications*, vol. 22, pp. 1349–1355, 2013.
- [18] G. Zhang, C. Zhang, and H. Zhang, "Improved k-means algorithm based on density canopy," *Knowledge-based systems*, vol. 145, pp. 289–297, 2018.
- [19] J. Wang and X. Su, "An improved k-means clustering algorithm," in *2011 IEEE 3rd international conference on communication software and networks*, pp. 44–46, IEEE, 2011.
- [20] C. Xiong, Z. Hua, K. Lv, and X. Li, "An improved k-means text clustering algorithm by optimizing initial cluster centers," in *2016 7th International Conference on Cloud Computing and Big Data (CCBD)*, pp. 265–268, IEEE, 2016.
- [21] C. Tang, J. Ding, H. Qi, and L. Zhang, "Smart forwarding deceptive jamming distribution optimal algorithm," *IET Radar, Sonar & Navigation*, 2024.
- [22] A. Mohammadi-Balani, M. D. Nayeri, A. Azar, and M. Taghizadeh-Yazdi, "Golden eagle optimizer: A nature-inspired metaheuristic algorithm," *Computers & Industrial Engineering*, vol. 152, p. 107050, 2021.
- [23] R. Tibshirani, G. Walther, and T. Hastie, "Estimating the number of clusters in a data set via the gap statistic," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 63, no. 2, pp. 411–423, 2001.
- [24] C. Z. Mooney, *Monte carlo simulation*. No. 116, Sage, 1997.